

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_

**Problem 1.** Find any local maximums/minimums of the following functions (hint: make a sign chart or use the second derivative).

**a)**  $f(x) = x^2$

**b)**  $f(x) = \frac{1}{2}x^2 - x + 3$

**c)**  $f(x) = x^3 - 1$

**d)**  $f(x) = 2x^3 + 3x^2 - 12x + 1$

*e)*  $f(x) = x^4 - 2x^2$

*f)*  $f(x) = \sin(x)$

*g)*  $f(x) = \cos^2(x)$

*h)*  $f(x) = e^{-x^3+3x}$